

CS 2601 Linear and Convex Optimization

11. Projected gradient descent

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Outline

- Algorithm and examples
- Convergence analysis

Projected gradient descent

Can we apply gradient descent to a constrained problem?

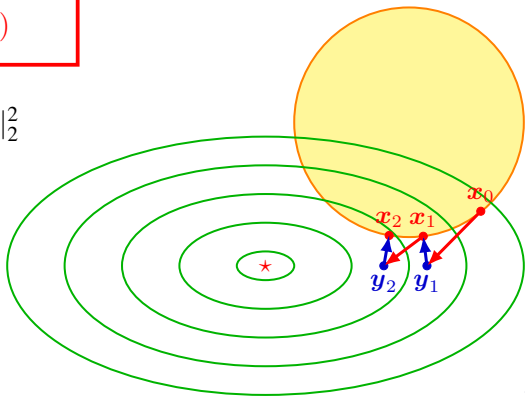
$$\min_{\mathbf{x} \in X} f(\mathbf{x})$$

What if $\mathbf{x}_k - t \nabla f(\mathbf{x}_k)$ is infeasible? Project it onto X !

$$\mathbf{x}_{k+1} = \mathcal{P}_X(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k))$$

where $\mathcal{P}_X(\mathbf{y}) = \operatorname{argmin}_{\mathbf{x} \in X} \|\mathbf{x} - \mathbf{y}\|_2^2$
is the projection of $\mathbf{y}$ onto X .

Useful if $\mathcal{P}_X$ can be computed efficiently.



Stopping criterion

Rewrite the projected gradient step as

$$\mathbf{x}_{k+1} = \mathbf{x}_k - t\mathbf{g}(\mathbf{x}_k), \quad \text{where } \mathbf{g}(\mathbf{x}) = \frac{1}{t}(\mathbf{x} - \mathcal{P}_X(\mathbf{x} - t\nabla f(\mathbf{x})))$$

Lemma. For convex f and X , $\mathbf{g}(\mathbf{x}^*) = \mathbf{0}$ iff $\mathbf{x}^*$ is a minimum of f over X .

Proof. Recall from slide 30 of §3, $\hat{\mathbf{x}} = \mathcal{P}_X(\mathbf{x})$ iff

$$\langle \mathbf{x} - \hat{\mathbf{x}}, \mathbf{z} - \hat{\mathbf{x}} \rangle \leq 0, \quad \forall \mathbf{z} \in X$$

so

$$\begin{aligned} \mathbf{g}(\mathbf{x}^*) = \mathbf{0} &\iff \mathbf{x}^* = \mathcal{P}_X(\mathbf{x}^* - t\nabla f(\mathbf{x}^*)) \\ &\iff \langle \mathbf{x}^* - t\nabla f(\mathbf{x}^*) - \mathbf{x}^*, \mathbf{z} - \mathbf{x}^* \rangle \leq 0, \quad \forall \mathbf{z} \in X \\ &\iff \langle \nabla f(\mathbf{x}^*), \mathbf{z} - \mathbf{x}^* \rangle \geq 0, \quad \forall \mathbf{z} \in X \\ &\iff \mathbf{x}^* \text{ is a minimum of } f \text{ over } X \end{aligned}$$

Note. $\mathbf{g}(\mathbf{x})$ plays a similar role as $\nabla f(\mathbf{x})$ does in gradient descent. We can stop when $\mathbf{g}(\mathbf{x}_k)$ is small, or equivalently when $\mathbf{x}_{k+1} - \mathbf{x}_k$ is small.

Examples

$\mathcal{P}_X$ can be efficiently computed for the following constraints.

- box constraint

$$\mathbf{a} \leq \mathbf{x} \leq \mathbf{b} \quad i.e. \quad a_i \leq x_i \leq b_i, i = 1, 2, \dots, n$$

Special case. ℓ_∞ constraint $\|\mathbf{x}\|_\infty \leq t$.

- affine constraint

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

- ℓ_2 constraint (e.g. ridge regression)

$$\|\mathbf{x}\|_2 \leq t$$

- quadratic constraint

$$\mathbf{x}^T \mathbf{Q} \mathbf{x} \leq 1$$

- ℓ_1 constraint (e.g. Lasso)

$$\|\mathbf{x}\|_1 \leq t$$

Projection onto box

For $X = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{a} \leq \mathbf{x} \leq \mathbf{b}\}$,

$$\begin{array}{ll}\min_{\mathbf{x}} & \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 \\ \text{s.t.} & \mathbf{a} \leq \mathbf{x} \leq \mathbf{b}\end{array}$$

The problem is decomposable,

$$\begin{array}{ll}\min_{x_i} & \frac{1}{2} (x_i - y_i)^2, \quad i = 1, 2, \dots, n \\ \text{s.t.} & a_i \leq x_i \leq b_i\end{array}$$

so

$$x_i = \begin{cases} a_i, & \text{if } y_i \leq a_i \\ y_i, & \text{if } a_i \leq y_i \leq b_i \\ b_i, & \text{if } y_i \geq b_i \end{cases}$$

or more compactly,

$$\mathcal{P}_X(\mathbf{y}) = \min\{\mathbf{b}, \max\{\mathbf{a}, \mathbf{y}\}\}$$

Projection onto affine space

For $X = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{A}\mathbf{x} = \mathbf{b}\}$, where $\mathbf{A} \in \mathbb{R}^{m \times n}$ with $\text{rank } \mathbf{A} = m$, $\mathbf{b} \in \mathbb{R}^m$.

Solve

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \mathbf{A}\mathbf{x} = \mathbf{b} \end{aligned}$$

By the Lagrange condition, there exists $\boldsymbol{\lambda} \in \mathbb{R}^m$ s.t.

$$\mathbf{x} - \mathbf{y} + \mathbf{A}^T \boldsymbol{\lambda} = \mathbf{0} \implies \mathbf{x} = \mathbf{y} - \mathbf{A}^T \boldsymbol{\lambda}$$

Plugging into $\mathbf{A}\mathbf{x} = \mathbf{b}$,

$$\mathbf{b} = \mathbf{A}\mathbf{y} - \mathbf{A}\mathbf{A}^T \boldsymbol{\lambda} \implies \boldsymbol{\lambda} = (\mathbf{A}\mathbf{A}^T)^{-1}(\mathbf{A}\mathbf{y} - \mathbf{b})$$

Thus

$$\mathcal{P}_X(\mathbf{y}) = \mathbf{y} - \mathbf{A}^T (\mathbf{A}\mathbf{A}^T)^{-1} (\mathbf{A}\mathbf{y} - \mathbf{b})$$

Projection onto ℓ_2 ball

For $X = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_2 \leq t\}$,

Solve

$$\begin{array}{ll}\min_{\mathbf{x}} & \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 \\ \text{s.t.} & \|\mathbf{x}\|_2^2 \leq t^2\end{array}$$

By the KKT conditions, there exists $\mu \geq 0$ s.t.

$$\mathbf{x} - \mathbf{y} + 2\mu\mathbf{x} = 0 \implies \mathbf{x} = \frac{\mathbf{y}}{1 + 2\mu} \propto \mathbf{y}$$

- If $\|\mathbf{y}\|_2 \leq t$, then $\mu = 0$ and $\mathbf{x} = \mathbf{y}$.
- If $\|\mathbf{y}\|_2 > t$, then $\mu > 0$ and $\mathbf{x} = \frac{t}{\|\mathbf{y}\|} \mathbf{y}$.

Thus

$$\mathcal{P}_X(\mathbf{y}) = \min \left\{ 1, \frac{t}{\|\mathbf{y}\|_2} \right\} \mathbf{y}$$

Projection onto ellipsoids

For $X = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x}^T \mathbf{Q} \mathbf{x} \leq 1\}$, where $\mathbf{Q} \succ \mathbf{O}$, need to solve

$$\begin{array}{ll} \min_{\mathbf{x}} & \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 \\ \text{s.t.} & \mathbf{x}^T \mathbf{Q} \mathbf{x} \leq 1 \end{array}$$

By the KKT conditions, there exists $\mu \geq 0$ s.t.

$$\mathbf{x} - \mathbf{y} + 2\mu \mathbf{Q} \mathbf{x} = 0 \implies \mathbf{x} = (\mathbf{I} + 2\mu \mathbf{Q})^{-1} \mathbf{y}$$

- If $\mathbf{y} \in X$, then $\mu = 0$ and $\mathbf{x} = \mathbf{y}$.
- If $\mathbf{y} \notin X$, then $\mu > 0$ and

$$1 = \mathbf{x}^T \mathbf{Q} \mathbf{x} = \mathbf{y}^T (\mathbf{I} + 2\mu \mathbf{Q})^{-1} \mathbf{Q} (\mathbf{I} + 2\mu \mathbf{Q})^{-1} \mathbf{y}$$

If $\mathbf{Q} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$ and $\mathbf{z} = \mathbf{U} \mathbf{y}$, then

$$\phi(\mu) = \mathbf{y}^T (\mathbf{I} + 2\mu \mathbf{Q})^{-1} \mathbf{Q} (\mathbf{I} + 2\mu \mathbf{Q})^{-1} \mathbf{y} = \sum_{i=1}^n \frac{\lambda_i z_i^2}{(1 + 2\mu \lambda_i)^2}$$

Projection onto ℓ_1 ball

For $X = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\|_1 \leq t\}$, need to solve

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 = \frac{1}{2} \sum_{i=1}^n (x_i - y_i)^2 \\ \text{s.t.} \quad & \|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i| \leq t \end{aligned} \tag{*}$$

$\mathcal{P}_X$ has no closed-form solution, but can be computed efficiently.

Observation 1. By symmetry, the general case can be reduced to the case $\mathbf{y} \geq \mathbf{0}$ by

$$\mathbf{x} = \text{sgn}(\mathbf{y}) \mathcal{P}_X(\text{abs}(\mathbf{y}))$$

e.g. if $(x_1, x_2) = \mathcal{P}_X(y_1, y_2)$, then $(-x_1, x_2) = \mathcal{P}_X(-y_1, y_2)$.

Observation 2. If $\mathbf{y} \geq \mathbf{0}$, then the solution $\mathbf{x} \geq \mathbf{0}$. If $x_i < 0$ for some i , then replacing x_i by $-x_i$ yields a better solution.

Projection onto ℓ_1 ball (cont'd)

Now focus on the case $\mathbf{y} \geq \mathbf{0}$. The problem reduces to

$$\begin{aligned} \min_{\mathbf{x}} \quad & \frac{1}{2} \sum_{i=1}^n (x_i - y_i)^2 \\ \text{s.t.} \quad & \sum_{i=1}^n x_i \leq t \\ & x_i \geq 0, \quad i = 1, 2, \dots, n \end{aligned}$$

By the KKT conditions, there exists $\mu_i \geq 0, i = 0, 1, \dots, n$ s.t.

$$x_i - y_i + \mu_0 - \mu_i = 0, \quad i = 1, 2, \dots, n$$

Thus

$$x_i = (y_i - \mu_0)^+$$

which is soft-thresholding with unknown μ_0 .

Projection onto ℓ_1 ball (cont'd)

μ_0 is determined by the constraint

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n (y_i - \mu_0)^+ \leq t$$

- If $\|\mathbf{y}\|_1 \leq t$, then $\mu_0 = 0$, $\mathbf{x} = \mathbf{y}$.
- If $\|\mathbf{y}\|_1 > t$, then $\mu_0 > 0$ and $\mathbf{x}$ can be found in a similar way as in the water filling solution.
 - ▶ Sort $\mathbf{y}$ s.t. $y_1 \geq y_2 \geq \dots \geq y_n$
 - ▶ Let

$$c_k = \frac{1}{k} \left(\sum_{i=1}^k y_i - t \right)$$

Then

$$\mu_0 = c_{k_0}$$

where

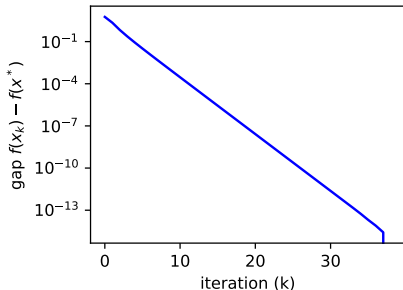
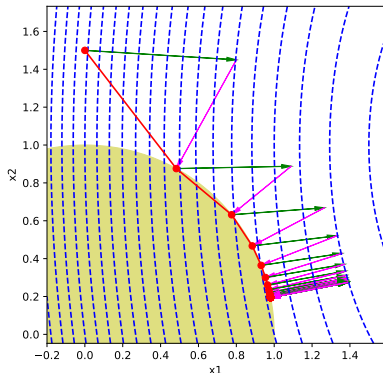
$$k_0 = \max\{k : c_k \leq y_k\}$$

Example: Ridge regression

$$\begin{aligned} \min_{\mathbf{w}} \quad & \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \|\mathbf{w}\|_2 \leq t \end{aligned}$$

$$\mathbf{X} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}, \quad t = 1$$

Step size $t = 0.1$, $\mathbf{w}_0 = (0, 1.5)$, $\mathbf{w}^* \approx (0.98100358, 0.19398965)$.

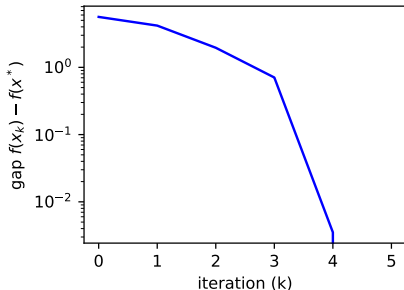
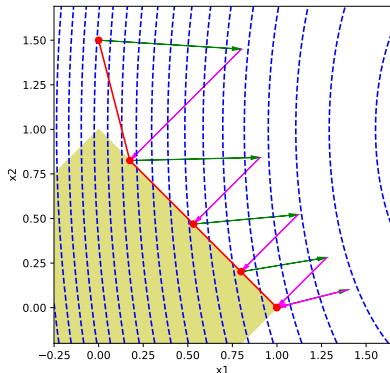


Example: Lasso

$$\begin{aligned} \min_{\mathbf{w}} \quad & \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \|\mathbf{w}\|_1 \leq t \end{aligned}$$

$$\mathbf{X} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}, \quad t = 1$$

Step size $t = 0.1$, $\mathbf{w}_0 = (0, 1.5)$, $\mathbf{w}^* = (1, 0)$.

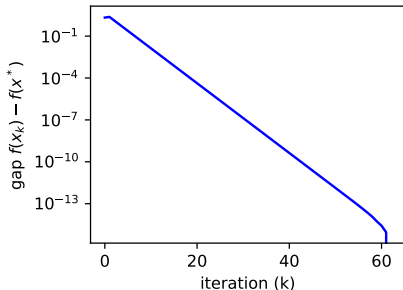
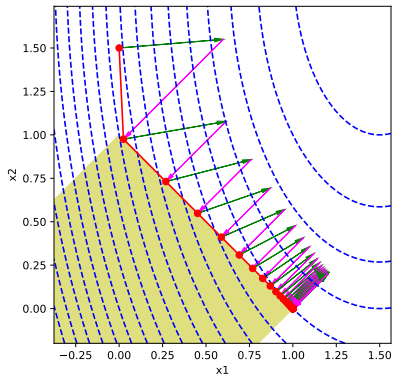


Example: Lasso

$$\begin{aligned} \min_{\mathbf{w}} \quad & \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \\ \text{s.t.} \quad & \|\mathbf{w}\|_1 \leq t \end{aligned}$$

$$\mathbf{X} = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 3 \\ 2 \\ 2 \end{bmatrix}, \quad t = 1$$

Step size $t = 0.1$, $\mathbf{w}_0 = (0, 1.5)$, $\mathbf{w}^* = (1, 0)$.



Outline

- Algorithm and examples
- Convergence analysis

Convergence analysis

Theorem. Let X be a nonempty convex set, and f an L -smooth convex function over X . Let $\mathbf{x}^*$ be a minimum of f over X . The sequence $\{\mathbf{x}_k\}$ produced by projected gradient descent with constant step size $t \leq \frac{1}{L}$ satisfies $f(\mathbf{x}_{k+1}) \leq f(\mathbf{x}_k)$ and

$$f(\mathbf{x}_k) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}^* - \mathbf{x}_0\|_2^2}{2tk}$$

If f is also m -strongly convex, then

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_2^2 \leq (1 - mt)\|\mathbf{x}_k - \mathbf{x}^*\|_2^2$$

and hence

$$\|\mathbf{x}_k - \mathbf{x}^*\|_2^2 \leq (1 - mt)^k \|\mathbf{x}_0 - \mathbf{x}^*\|_2^2$$

Note. The results are similar to those for proximal gradient descent (slide 13 of §8). In fact, we will see that projected gradient descent can be considered as a special case of proximal gradient descent.

Connection to proximal gradient descent

Define the **indicator** I_X of a set X by

$$I_X(\mathbf{x}) = \begin{cases} 0, & \text{if } \mathbf{x} \in X \\ +\infty, & \text{if } \mathbf{x} \notin X \end{cases}$$

Note. I_X is a convex function iff X is a convex set.

The proximal operator for I_X is simply the projection onto X ,

$$\begin{aligned} \text{prox}_{I_X}(\mathbf{y}) &= \underset{\mathbf{x}}{\operatorname{argmin}} \left\{ \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 + I_X(\mathbf{x}) \right\} \\ &= \underset{\mathbf{x} \in X}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_2^2 \\ &= \mathcal{P}_X(\mathbf{y}) \end{aligned}$$

Connection to proximal gradient descent (cont'd)

Note

$$\min_{\mathbf{x} \in X} f(\mathbf{x}) \iff \min_{\mathbf{x}} \{f(\mathbf{x}) + I_X(\mathbf{x})\}$$

and

$$\begin{aligned}\mathbf{x}_{k+1} &= \mathcal{P}_X(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k)) \\ &= \text{prox}_{I_X}(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k)) \\ &= \text{prox}_{t_k I_X}(\mathbf{x}_k - t_k \nabla f(\mathbf{x}_k))\end{aligned}$$

where the last step uses $I_X(\mathbf{x}) = t_k I_X(\mathbf{x})$ for $t_k > 0$.

Projected gradient descent for $\min_{\mathbf{x} \in X} f(\mathbf{x})$ is the same as proximal gradient descent for $\min_{\mathbf{x}} \{f(\mathbf{x}) + I_X(\mathbf{x})\}$!

The convergence analysis for proximal gradient descent applies to projected gradient descent with $h(\mathbf{x}) = I_X(\mathbf{x})$.